

Javier Herrero Pérez

tlf: 656181225 | jherrerop88@gmail.com | www.linkedin.com/in/javier-herrero-pérez | github.com/hepejaAED | hepejaaed.github.io

About me

I hold a degree in Physics from the University of Salamanca, with a strong foundation in mathematical modelling, statistics and data analysis. I am currently pursuing a Master's degree in Data Science at the University of Valencia and have hands-on experience in applied AI, including machine learning and natural language processing, gained during an internship at the AIR Institute.

I am particularly interested in Generative AI and LLM-based systems, and I am seeking junior Data Scientist roles where I can contribute to building and deploying data-driven solutions in real-world products.

Technical Skills

Programming: Python, SQL, R, Fortran

Data & ML Libraries: NumPy, Pandas, Scikit-learn

Deep Learning Frameworks: TensorFlow, PyTorch, Keras

MLOps & Tools: Git, AWS, Jupyter Notebook, MongoDB, Cassandra, LaTeX

Languages: Spanish (native), English (B1+: Professional communication and technical proficiency)

Experience

Data Scientist | Internship

Jul 2024 – Aug 2024

AIR Institute (International Centre for Artificial Intelligence Research)

Spain

- **Workplace Risk Classification:** Took a real sensor dataset from scratch and put together a 6-class risk detection pipeline: feature engineering (temporal exposure, exponential decay weighting), SMOTE to handle a 27:1 imbalance, and a Random Forest that hit **100% accuracy and macro F1 = 0.99** on a stratified 80/20 split. Used SHAP to explain the model and show that CO₂ and cumulative exposure time were the dominant drivers. Also trained an XGBoost regressor to predict CO₂ from environmental sensors, hitting **R² = 0.9999 and MAE = 27.74 ppm** (<6% error).
- **Candidate Profiling (NLP):** Built a feature extraction pipeline with Whisper and PySentimiento to pull tone, emotion and linguistic signals from candidate interviews; the features fed a shortlisting classifier and measurably improved its precision and recall.

Projects

Clickbait Detection in News Headlines

Developed an NLP model to detect clickbait using cosine similarity and topic modeling (Python, Scikit-learn, Transformers).

Calorie Estimation from Food Images

Built a deep learning pipeline to estimate calories from images using CNN models (Python, TensorFlow, Hugging Face).

Exploratory Analysis of Labour Market Data

Performed EDA on Spanish labour market data to identify gender inequality patterns (R, tidyverse, ggplot2).

Education

Master's in Data Science

Sept 2025 - Jun 2026 (expected)

University of Valencia

Spain

- GPA: 8.86/10

Bachelor's Degree in Physics

Sept 2021 - Jul 2025

University of Salamanca

Spain

- GPA: 7.25/10